

4. Viewing and Understanding Data in Pandas

When we load data into Pandas, first we need to:

- View data
- Understand structure
- Check rows and columns
- Check data types
- Find missing values

Pandas provides useful functions for this.

First Step: Import Pandas

```
import pandas as pd
```

Sample Data

We will use this DataFrame for all examples.

```
data = {  
    "Name": ["Rahul", "Aman", "Priya", "Neha", "Rohit"],  
    "Age": [20, 21, 19, 22, 23],  
    "Marks": [85, 90, 88, 75, 95],  
    "City": ["Delhi", "Patna", "Mumbai", "Kolkata", "Pune"]  
}  
  
df = pd.DataFrame(data)  
  
print(df)
```

Output

	Name	Age	Marks	City
0	Rahul	20	85	Delhi

1	Aman	21	90	Patna
2	Priya	19	88	Mumbai
3	Neha	22	75	Kolkata
4	Rohit	23	95	Pune

1. head()

head() shows first rows of data.

Useful when:

- Dataset is large
 - Want quick preview
-

Syntax

df.head()

Example 1: Default head()

```
print(df.head())
```

Shows first 5 rows.

Output

	Name	Age	Marks	City
0	Rahul	20	85	Delhi

1	Aman	21	90	Patna
2	Priya	19	88	Mumbai
3	Neha	22	75	Kolkata
4	Rohit	23	95	Pune

Example 2: Show First 2 Rows

```
print(df.head(2))
```

Output

	Name	Age	Marks	City
0	Rahul	20	85	Delhi
1	Aman	21	90	Patna

2. tail()

tail() shows last rows.

Useful for:

- Checking end data
- Verifying latest records

Syntax

```
df.tail()
```

Example 1: Default tail()

```
print(df.tail())
```

Shows last 5 rows.

Example 2: Last 2 Rows

```
print(df.tail(2))
```

Output

	Name	Age	Marks	City
3	Neha	22	75	Kolkata
4	Rohit	23	95	Pune

3. shape

shape tells:

- Number of rows
 - Number of columns
-

Syntax

```
df.shape
```

Example

```
print(df.shape)
```

Output

(5, 4)

Understanding Output

(rows, columns)

So:

Value	Meaning
5	Total rows
4	Total columns

4. columns

columns shows all column names.

Syntax

```
df.columns
```

Example

```
print(df.columns)
```

Output

```
Index(['Name', 'Age', 'Marks', 'City'], dtype='object')
```

Convert into List

```
print(list(df.columns))
```

Output

```
['Name', 'Age', 'Marks', 'City']
```

5. info()

info() gives complete dataset information.

It shows:

- Total rows
- Column names

- Data types
- Missing values
- Memory usage

Syntax

```
df.info()
```

Example

```
print(df.info())
```

Output Example

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 5 entries, 0 to 4
```

Data columns (total 4 columns):

Name 5 non-null object

Age 5 non-null int64

Marks 5 non-null int64

City 5 non-null object

dtypes: int64(2), object(2)

Understanding info()

Part	Meaning
RangeIndex	Total rows
Data columns	Total columns
non-null	Non-empty values
dtypes	Data types

6. describe()

describe() gives statistical summary.

Works mainly on numeric columns.

Syntax

```
df.describe()
```

Example

```
print(df.describe())
```

Output Example

	Age	Marks
count	5	5
mean	21	86.6
std	1.58	7.3
min	19	75
25%	20	85
50%	21	88
75%	22	90
max	23	95

Understanding describe()

Term	Meaning
count	Total values
mean	Average
std	Standard deviation
min	Minimum value
max	Maximum value

25%	First quartile
50%	Median
75%	Third quartile

7. dtypes

dtypes shows data type of each column.

Syntax

df.dtypes

Example

```
print(df.dtypes)
```

Output

```
Name    object
Age      int64
Marks    int64
City     object
dtype: object
```

Understanding Data Types

Data Type	Meaning
int64	Integer number
float64	Decimal number
object	Text/String
bool	True/False

Real-Life Example

Suppose company has employee dataset.

Before analysis:

1. Check first rows → head()
2. Check last rows → tail()
3. Check size → shape
4. Check columns → columns
5. Check missing values → info()
6. Check statistics → describe()
7. Check data types → dtypes

Full Example Together

```
import pandas as pd

data = {
    "Name": ["Rahul", "Aman", "Priya"],
    "Age": [20, 21, 19],
    "Marks": [85, 90, 88]
}

df = pd.DataFrame(data)

print(df.head())

print(df.tail())

print(df.shape)

print(df.columns)

print(df.info())

print(df.describe())

print(df.dtypes)
```

Summary Table

Function	Purpose
head()	Show first rows
tail()	Show last rows
Shape	Rows and columns count
Columns	Column names
info()	Dataset information
describe()	Statistical summary
Dtypes	Data types